Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad Nacional del Sur
Short name / acronym used in projects	UNS
Participant Identification Code (PIC), if available	PIC de CONICET 998619754 (to be confirmed)
Website(s)	https://inibibb.conicet.gov.ar/
Country	Argentina
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Camino La Carrindanga Km7, Edificio E1, 8000, Bahía Blanca, Argentina.
Department(s), institute(s), research group(s) or unit(s) that would be involved	Lipid and Signal Transduction Laboratory, Instituto de Investigaciones Bioquímicas de Bahía Blanca (INIBIBB), Consejo Nacional de Investigaciones Científicas y Tecnologicas (CONICET), Universidad Nacional del Sur (UNS). *(The INIBIBB is research institute affiliated with both, CONICET and UNS).* https://inibibb.conicet.gov.ar/laboratorio-de-lipidos-y-transduccion-de-senales/
Additional institutional resources relevant to collaboration	Facilities and equipment of the INIBIBB-CONICET-UNS: https://inibibb.conicet.gov.ar/equipos/ Animal Facilities: CONICET–UNS Absolute Barrier Animal Facility. https://inibibb.conicet.gov.ar/bioterio/ Other services available at UNS: Flow cytometry services. https://inbiosur.conicet.gov.ar/citometria-de-flujo/

Main contact person for this profile	**Melina Valeria Mateos,** Independent Researcher INIBIBB-CONICET. Associate Professor, Pharmacology and Toxicology, Department of Biology, Biochemistry and Pharmacy, UNS.
Main contact email	mvmateos@criba.edu.ar melinavaleriamateos@gmail.com
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	X Retina
☐ Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Retinal pigment epithelium, retina, inflammation, oxidative stress, lipid signaling, pre-clinical models
Geographies of interest for exchanges (optional)	Spain, Germany, France

Overview of your research / innovation focus
Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words). **Lipid and Signal Transduction Lab at INIBIBB (PI: Melina V. Mateos, Members: Vicente Bermúdez, María Sol Echevarría, Paula E. Tenconi).** Our overall goal is to investigate novel molecular mechanisms that may modulate the inflammatory process in ocular and retinal degenerative diseases that can lead to vision loss, including uveitis, dry eye disease, age-related macular degeneration and diabetic retinopathy, among others. The pharmacological therapeutic options currently available for these inflammatory ocular diseases remain limited, that is why it is essential to deepen our understanding of the underlying molecular mechanisms involved in order to identify novel therapeutic targets. Over the past 13 years, our laboratory has characterized the role of the Phospholipase D (PLD) signaling pathway, which generates bioactive lipid second messengers, in the inflammatory response of retinal pigment epithelium (RPE) cells exposed to inflammatory injury. Pharmacological inhibition of the PLD pathway not only prevented the inflammatory response and oxidative stress but also preserved RPE cell function and viability. These findings position classical PLD isoforms (PLD1 and PLD2) as potential therapeutic targets to simultaneously counteract and/or prevent the inflammatory response and oxidative stress, two key factors in the pathophysiology of the above mentioned inflammatory retinal diseases. Currently, we are investigating the effects of PLD inhibitors in an *in vivo* model of uveitis as well as in an *ex vivo* model of neurodegeneration using porcine neural retinal explants. Our preliminary data from the endotoxin-induced uveitis rat model show that the PLD2 inhibitor is able to reduce clinical signs, total protein concentration in the aqueous humor and cellular infiltration in both the anterior and posterior segments of the eye. As future projects, we aim to investigate the role of the PLD pathway in ocular surface diseases, such as dry eye disease, and in the physiopathology of retinoblastoma (RB).

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our approach may lead to new opportunities for the development of safer and more effective pharmacological therapies for inflammatory ocular diseases. In the future, the implementation of innovative therapeutic strategies could not only significantly improve patients' quality of life but also contribute to strengthening public health systems. Regarding RB, in Argentina is often diagnosed at a later stage, and survival rates are lower in the less developed regions. As a result, these patients may progress to more aggressive forms of RB and have a higher probability of recurrence or resistance to conventional therapy. Therefore, despite major advances in treatment, these cases remain difficult to manage with currently available pharmacological options, and most require enucleation. In this context, the search for new compounds with therapeutic activity against retinoblastoma has become a priority.

Our laboratory has extensive expertise in a wide range of cell and molecular biology techniques, as well as microscopy. All members of the laboratory have completed training at UNS (equivalent to a FELASA Function A and B of the Functions framework defined in EU Directive 2010/63/EU) for working with laboratory animals. We have experience in both basic and preclinical research, using *in vitro* cell culture models and the endotoxin-induced uveitis (EIU) rat model. We have the infrastructure and trained personnel required to establish and implement several animal models using specific pathogen-free (SPF) rodents. We are currently establishing a rat model of dry eye disease and in the near future we will be implementing behavioral assays to assess visual function, including the visual cliff test and the looming test.

In addition to the members of our laboratory, the team participating in this consortium includes researchers from other laboratories at INIBIBB whose research lines, while not directly related to vision science, provide expertise that we believe will make a valuable contribution to the consortium:

-**Dr. Gerardo Martín Oresti,** together with his principal collaborator, **Dr. Florencia Santiago Valtierra**, investigate **lipid metabolism** and membrane remodeling during cellular differentiation, with particular emphasis on polyunsaturated fatty acids (PUFAs), glycerophospholipids, and highly unsaturated sphingomyelin and ceramide species containing very-long-chain PUFAs. Their research combines *in vivo*, *ex vivo*, and *in vitro* models with chromatographic (GC-MS and HPLC), biochemical, and molecular approaches.

-**Dr. Daniel Alejandro Peñalva** has expertise in assessing the **biophysical properties of membranes**. His expertise includes the development and study of model membrane systems, such as liposomes, giant unilamellar vesicles, monolayers, and supported lipid bilayers, and the assessment of the biophysical properties of both biological and model membranes.

Our team has extensive experience primarily in the development and management of preclinical models. Given our background in biochemistry, molecular biology and biophysics, together with our expertise in describing molecular mechanisms, we can contribute with scientific rigor to the preclinical stages of research projects, supporting their progression toward translational applications. Our principal added value lies in the integration of ocular preclinical models with lipid signaling, lipidomic and membrane biophysics approaches. We can provide mechanistic and functional validation of candidate therapeutic targets in RPE cells, retinal explants and rodent models of ocular inflammatory diseases.

In summary the team can contribute to:

- preclinical evaluation of anti-inflammatory and neuroprotective compounds
- lipid and fatty-acid profiling of ocular tissues, cells and biological fluids
- characterization of membrane remodeling associated with oxidative stress and inflammation
- development of lipid-based biomarkers and pharmacodynamic readouts
- training in experimental design, animal models, lipid analysis and membrane biophysics

These capabilities complement clinical, organoid, imaging, drug-delivery and patient-oriented partners by providing a mechanistic and preclinical bridge between target discovery and translational validation.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

-Molecular and cell biology: design, development, optimization, and execution of cell and molecular biology techniques (evaluation of cell viability, cell cycle, gene expression at the RNA and protein levels, oxidative stress, and inflammatory responses). Evaluation of metabolic pathways and enzyme activities using radiolabeled substrates (^{3}H, ^{14}C, ^{32}P). The INIBIBB is authorized by Argentina's Nuclear Regulatory Authority (ARN) to handle radioisotopes.

-Lipid biochemistry analyses: lipid and fatty-acid analysis using gas chromatography, GC–MS, HPLC, thin-layer chromatography, and complementary molecular techniques. Quantitative and qualitative analysis of lipids and saturated, monounsaturated, and polyunsaturated fatty acids (omega-3, omega-6, and omega-9), including both *cis* and *trans* isomers. Lipid and fatty-acid profiling, data analysis and interpretation.

- Analysis of membrane biophysical properties: study of model membrane systems, such as liposomes, giant unilamellar vesicles, monolayers, and supported lipid bilayers.

- Laboratory animal experimentation conducted by specialized technical and professional staff: Animal handling, restraint, identification, and sex determination techniques. Substance administration and sample collection procedures. Design and implementation of sedation, analgesia, and anesthesia protocols. Performance of surgical procedures. Conduct of euthanasia procedures and necropsies. Monitoring of the genetic quality of laboratory animals: sample collection and genotyping using molecular biology techniques. Implementation of specialized diets and oral administration of substances through drinking water. Performance of behavioral tests in dedicated rooms equipped for proper execution, including behavioral monitoring and video recording systems.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

INIBIBB offers well-established infrastructure and recognized expertise in biochemical research.

Facilities and equipment of the INIBIBB-CONICET-UNS: https://inibibb.conicet.gov.ar/equipos/

-Four cell culture rooms, equipped with laminar flow hoods, incubators, water baths, light microscopes, centrifuges, liquid nitrogen tanks.
-80°C and -20°C freezers.
-Centrifuges and ultracentrifuge
- Cryostat microtome (Leica CM1860)
-RT-PCR
- G:BOX Chemi XRQ imaging system with data processing software
-GC/MS Gas Chromatograph (Agilent 8890/5977C GC/MSD system. Configuration: Split/splitless (S/SL) injection port and a single quadrupole mass spectrometer equipped with a diffusion pump).
-High-Performance Liquid Chromatograph (HPLC). Agilent 1260 Infinity with a UV detector.
- Inverted fluorescence microscope: Nikon Eclipse TE2000-S equipped with a high-sensitivity monochrome digital camera (Nikon DS-Qi2 model).
-Confocal Microscope. LSM 900 Confocal Microscope with two confocal detectors and an Axio Observer 7 inverted microscope.
-Multiskan GO microplate reader (UV/visible spectrofluorometer).
-Spectrofluorometers (Agilent Cary Eclipse and AMINCO SLM 4800)
-Equipment for the formation of Giant Unilamellar Vesicles (Vesicle Prep Pro, Nanion Technologies GmbH)
-Extrusion system for the generation of Large Unilamellar Vesicles (LiposoFast, Avestin Inc.).
Animal Facilities: CONICET–UNS Absolute Barrier Animal Facility. Production and supply of SPF rodents: C57BL/6 mice, BALB/c mice, and Wistar rats. Rodent housing in individually ventilated cage (IVC) racks and conventional cages. Laboratory animal experimentation services provided by specialized technical and professional staff.
https://inibibb.conicet.gov.ar/bioterio/
Other services available at UNS: Flow cytometry services. https://inbiosur.conicet.gov.ar/citometria-de-flujo/

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Inbound staff could acquire advanced knowledge and practical expertise in multidisciplinary biomedical research approaches, integrating molecular and cell biology, lipid biochemistry, membrane biophysics, and *in vitro* and *in vivo* experimental models. They would learn how to design, optimize, and perform experimental strategies to investigate cellular and molecular mechanisms, including the evaluation of cell viability, cell cycle regulation, gene and protein expression, oxidative stress, inflammatory responses, metabolic pathways and signal transduction, and enzymatic activities.
They could also develop expertise in lipid and fatty-acid profiling, including the analysis and interpretation of lipid composition and metabolism, as well as the study of how lipid alterations influence membrane structure and function through model membrane systems.
The INIBIBB was established in 1970 at the UNS. Over the years, INIBIBB has become a leading center for research in Biochemistry and Neuroscience. Committed to both scientific research and academic training, the Institute continues to contribute to the advancement of knowledge in these fields. Additionally, our laboratory has more than 13 years of experience conducting research in vision sciences.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Through participation in this consortium, our team will benefit from access to cutting-edge technologies, multidisciplinary expertise, and innovative translational strategies that will enhance our research capabilities and expand our current research in retinal diseases and visual health. We are particularly interested in learning about induced pluripotent stem cell (iPSC) technology and retinal organoids.
Furthermore, this program will provide valuable opportunities for knowledge exchange, collaborative research, and capacity building, enabling the implementation of new methodologies. We have also a strong interest in establishing new collaborations as well as in strengthening the ongoing ones.
Since our research line is predominantly focused on basic science, we are interested in gaining knowledge of the regulatory, legal, and patient-related aspects involved in clinical trial design. This knowledge will be essential to support the future translation of our findings from fundamental research into clinically relevant applications.
Participation in this program will make a significant contribution to the training of PhD students and postdoctoral fellows, enabling them to learn new methodologies and, above all, giving them the opportunity to gain research and academic experience in another country. For senior researchers the exchange program would give them the opportunity to acquire and implement novel methodologies, analytical tools, and experimental strategies that could be implemented at our institution. In addition, the program will also provide training opportunities for faculty members at UNS, which will greatly enrich the university's graduate academic programs.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Melina Valeria Mateos	Independent Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	mvmateos@crib.edu.ar melinavaleriamateos@gmail.com
Vicente Bermúdez	Assistant Researcher	INIBIBB-CONICET-UNS	Researcher	Experienced	bermudez@inibibb-conicet.gob.ar
Gerardo Martín Oresti	Independent Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	gmoresti@criba.edu.ar

Daniel Alejandro Peñalva	Associate Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	dpenalva@criba.edu.ar
María Sol Echevarría	PhD Student	INIBIBB-CONICET-UNS	Researcher	Early-stage	msechevarria@inibibb-conicet.gob.ar
Paula Estefanía Tenconi	Academic staff member	UNS	Researcher	Early-stage	petenconi@inibibb-conicet.gob.ar
TBD	2 x PhD students	INIBIBB-CONICET-UNS	Researcher	Early-stage	

Main disciplines represented in your team	Biochemistry, cell biology, molecular biology, signal transduction, biochemistry of lipids, biophysics
Experience in EU / international / multi-partner collaborative projects	Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

1. “Shielding retinal pigment epithelium cells from high glucose-induced oxidative stress: the protective effect of Phospholipase D (PLD) pathway inhibition”. Echevarría, MS; Tenconi, PE; Bermúdez, V; Calandria, JM; Bazán, NG, Mateos, MV*. *Biochimica et Biophysica Acta- Molecular Cell Research* (2026) 1873, 120121. (SCIMAGO: Q1). doi: 10.1016/j.bbamcr.2026.120121.

2. “Targeting Phospholipase D Pharmacologically Prevents Phagocytic Function Loss of Retinal Pigment Epithelium Cells Exposed to High Glucose Levels”. Bermúdez, V; Tenconi, PE; Echevarría, MS; Asatrian, A; Calandria, JM; Giusto, NM; Bazan, NG; Mateos, MV*. *International Journal of Molecular Sciences* (2022) 23, 11823. (SCIMAGO: Q1). doi: 10.3390/ijms231911823.

3. “New Retinal Pigment Epithelial Cell Model to Unravel Neuroprotection Sensors of Neurodegeneration in Retinal Disease”. Asatryan, A; Calandria, JM; Kautzmann, MA; Jun, B; Gordon, WC; Do, K; Bhattacharjee, S; Pham, TL; Bermúdez, V; Mateos, MV; Heap, J and Bazan, NG. *Frontiers in Neuroscience* (2022) 16-926629. (SCIMAGO Q2). doi: 10.3389/fnins.2022.926629. eCollection 2022.

4. "High glucose-induced phospholipase D activity in retinal pigment epithelium cells: new insights into the molecular mechanisms of diabetic retinopathy". Tenconi, PE; Bermúdez, V; Oresti, GM; Giusto, NM; Salvador, GA and Mateos, MV*. *Experimental Eye Research* (2019) 184, 243-257. (SCIMAGO: Q1 Ophthalmology). doi: 10.1016/j.exer.2019.04.028.

5. “The Phospholipase D pathway mediates the inflammatory response of the retinal pigment epithelium”. Mateos, MV*; Kamerbeek, CB; Giusto, NM and Salvador, GA. *International Journal of Biochemistry & Cell Biology* (2014) 55, 119-128. (SCIMAGO: Q1). doi: 10.1016/j.biocel.2014.08.016.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). *Indicate if international.*

National grants:

- -PIP 2021-2023 (11220200101062CO) granted by National Scientific and Technical Research Council (CONICET), Argentina. Period December 2021- December2025. Project title: "Classic PLD isoforms as potential therapeutic targets for retina inflammatory pathologies" (Role PI).
  -PGI 24/B368 granted by Secretaría General de Ciencia y Tecnología, Universidad Nacional del Sur, Argentina (SGCyT-UNS). Period January 2025-December 2028. Project title: "Canonical Phospholipase D (PLD) isoforms as potential therapeutic targets for uveitis and inflammatory diseases of the ocular surface"" (Role PI).

International collaborations and scientific networks:

- Thanks to a ***2018 Developing Country Eye Research Fellowship (DCERF)*** (granted by ARVO Foundation to Melina Mateos) and to a ***2019 ARVO Foundation Collaborative Research Fellowship*** (granted to Vicente Bermúdez) our group began a collaboration with Dr. Nicolás Bazán's laboratory (Neuroscience Center of Excellence, School of Medicine, Louisiana State University Health New Orleans, New Orleans, USA). During this collaboration, we contributed to the characterization of a novel human retinal pigment epithelium (RPE) cell line, designated ABC cells, which exhibits phenotypic, genetic, and functional traits closely resembling native RPE. Also, we were able to study the role of the PLD pathway in this novel experimental model. As a result of this collaboration, we have published 3 research papers outlined in the previous section.
- Thanks to a ***2023 ARVO Foundation Collaborative Research Fellowship*** (granted to Paula Estefanía Tenconi) our laboratory began a collaboration with Dr. Wai Kit Chu (Deparment of Ophthalmology and Visual Sciences, The Chinese University of Hong Kong). This collaboration enabled us to investigate the effects of PLD inhibitors in the endotoxin-induced uveitis rat model and to initiate the PLD-retinoblastoma project.
- During 2025, an ***International Union of Biochemistry and Molecular Biology (IUBMB) Fellowship*** awarded to María Sol Echevarría, a doctoral fellow in our laboratory, enabled her to complete a research stay in Dr. Iván Fernández Bueno's laboratory (IOBA, Universidad de Valladolid, Spain), where she was trained in the porcine neuroretinal explant model. This fellowship marked the beginning of a new collaboration for our laboratory.
- From May 2026 to May 2028, **Melina Mateos serves as President of the Argentine Association for Vision and Ophthalmology Research (AIVO)**. AIVO is the international chapter affiliated with ARVO (https://www.arvo.org/members/international-chapter-affiliates/aivo) and is an interdisciplinary scientific society composed of ophthalmologists, biochemists, biologists, pharmacologists, and bioengineers, among others, engaged in research on vision and ophthalmology. AIVO also works in close collaboration with the Argentine Council of Ophthalmology (CAO), the Argentine Society for Ocular Surface (SASO), the Argentine Society of Ophthalmology (SAO) and the Argentine Society for Research in Biochemistry and Molecular Biology (SAIB).

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
None.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Visa processing, if required.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. • Preclinical evaluation of anti-inflammatory, anti-oxidants and neuroprotective compounds: • Lipid and fatty-acid profiling of ocular tissues, cells and biological fluids: lipid and fatty-acid analysis using gas chromatography, GC–MS, HPLC, thin-layer chromatography, and complementary molecular techniques. Characterization of membrane remodeling associated with oxidative stress and inflammation • *In vitro* experimental design, *in vivo* animal models.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. • Access to state-of-the-art infrastructures and specialized technologies beyond the current capabilities of our institute. This possibility would significantly strengthen our research capacity (advanced imaging platforms, omics technologies, high-throughput screening approaches, AI-driven analytical tools). • Acquire training in advanced human retinal modelling using iPSC technology. • Acquire training in organotypic neural retina culture platform. • *In vivo* preclinical models of retinal disease: acquire expertise in animal models (mainly rodent models) of ocular diseases, with the aim of subsequently implementing and further developing these models at INIBIBB. • Acquire expertise in electroretinography for the evaluation of visual function, enabling the implementation of these approaches at INIBIBB. • Development of long-term international collaboration frameworks to build durable partnerships, supported by joint research programs, funding applications, and the generation of high-impact co-authored scientific outputs. • Exchange of expertise and methodologies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None.

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organization / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organization. Submission of this form confirms that the contributor agrees to this internal use. Organizations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Melina Valeria Mateos
Date completed: July 18, 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)